Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/23/2004
PI Name:	Magda, Jules Ph.D.		
Project Title:	Novel Microstructures for Polymer-Liquid Crystal Composite Materials		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Polymers		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jj.magda@m.cc.utah.edu	**Fax:**	FY 801 585-9291
PI Organization Type:	UNIVERSITY	**Phone:**	801 581-7536
Organization Name:	University of Utah		
PI Address 1:	Room 3290, 50 South Central Campus Drive		
PI Address 2:	Department of Chemical and Fuels Engineering		
PI Web Page:			
City:	Salt Lake City	**State:**	UT
Zip Code:	84112-9203	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2000	**End Date:**	11/01/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sibille, Laurent	**Contact Phone:**	256 544-5221
Contact Email:	laurent.sibille@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smith, G. (University of Utah) Glaser, M. (University of Colorado) Clark, N. (University of Colorado) Radzihovsky, L. (University of Colorado) Bowman, C. (University of Colorado) Bellini, T. (University of Milan)		
Grant/Contract No.:	NAG8-1665		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Liquid Crystal Dispersed Colloids (LCDCs) are a new type of composite material, consisting of colloidal polymer particles dispersed in a liquid crystal (LC) matrix. Thermotropic-Lytropic Liquid Crystals (TLLCs) are a hybrid of thermotropic and lyotropic liquid crystalline systems that may potentially combine the useful features of both types of self-assembling materials at the nanoscale level. Interfacial properties are crucial to the overall properties of both LCDCs and TLLCs. In order to understand the LC interface, we are developing molecular-based computer models of important LC molecules. The molecular models will be validated by comparisons of computer simulation results to experimental measurements of the liquid crystal interfacial tension. A model TLLC consisting of a microemulsion of aqueous droplets dispersed in an LC matrix is also being studied via microscopy and light scattering. The LC microemulsion may also be used in highly sensitive chemical sensors used to detect surface active analytes that reversibly bind at the liquid crystal interface. To explore the novel phases of LCDCs and TLLCs, and to develop the ability to predict the phases using molecular models and interfacial tension measurements. The novel materials to be prepared may be useful as optical switches in optical telecommunications networks, and could also be used in highly sensitive chemical sensors used to detect surface active analytes or to separate surface active molecules on the basis of chirality. When used in chemical sensors, the novel materials could be designed to change color or transparency as a signal that the chemical of interest is present in the environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Novel materials prepared from TLLCs or LCDCs are likely to have properties unlike those of any other material previously studied, with potential application in sensors, optical communications, and membrane separations.
Task Progress:	Have developed a coarse-grained molecular model that can predict the interfacial properties of smectic liquid crystals. Have fabricated and studied the phase diagram of a TLLC consisting of nanosize reverse micelles or droplets dispersed in an LC medium consisting of nematic liquid crystal 5CB. Have developed a model that predicts the phase diagram of this TLLC. Have measured the interfacial tension between water and 5CB using pendant drop methods.
Bibliography Type:	Description: (Last Updated: 03/28/2005)
Abstracts for Journals and Proceedings	Magda, J. J.; Smith, G. D.; Tian, Pu; Ramanathan, K.; Kim, H.; Glaser, M. A.; Clark, N. A.; Bowman, C.; Radzihovsky, L.; Nagel, S.; Kang, D.; Mantegazza, F.; Bellini, T.; Caggioni, M. "Novel microstructures for polymer-liquid crystal composite materials" 2002 Microgravity Materials Science Conference, Huntsville, AL NASA Conference Publication 2003-212339, 383-389 , Jan-2003
Articles in Peer-reviewed Journals	Kim, J.-W.; Kim, H.; Lee, M.; Magda, J.J. "Interfacial Tension of Nematic Liquid Crystal/Water Interface with Homeotropic Surface Alignment" Langmuir , Jul-2004
Articles in Peer-reviewed Journals	Bellini, T.; Caggioni, M.; Clark, N.A.; Mantegazza, F.; Maritan, A.; Pellizzola, A. "Fluctuation Mediated Interaction and Phase Separation of Nanoparticles in a Liquid Crystal Solvent" Physical Review Letters 91(8), 085704/1-085704/4 , Jan-2003